

Shreyaa Sanjay

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EDUCATION

Cornell University

Bachelor of Science in Computer Science, Minor in Electrical and Computer Engineering

Ithaca, NY

Expected May 2028

Relevant Coursework: Functional Programming in OCaml, Data Structures and OOP, Full-Stack Web Development, Machine Learning, Discrete Math, Linear Algebra, Differential Equations, Digital Logic and Embedded Systems in Verilog

Campus Involvement: Resident Advisor, Cornell Hack4Impact, Women in Computing at Cornell

EXPERIENCE

CS3 (Center for Smart StreetScapes, WINLAB)

Software Engineering Intern, Sensing and Reasoning Team

New Brunswick, NJ

May 2026 – August 2026

- Built multi-agentic stack with **8** engineers to answer natural-language questions about a live **4-camera** smart space
- Designed infra for LLM agents to autonomously plan workflows, generate and execute code, and orchestrate tool calls
- Replaced manual footage review and sensor querying, achieved **87.8% accuracy across 150+ automated unit tests**
- Integrated Tracefix (CS3 TLA+ Verification Protocol) to validate **120+** agent workflows, repaired **23** invalid protocols
- Built **Docker harness** to execute agent-written code and a **cache** layer that classifies user intent with a template protocol before deploying agents, reducing end-to-end runtime to **20–30 seconds**

Cornell Electric Vehicles

Software Engineer, 1st Place Data & Telemetry Shell-Eco Marathon ([linked technical paper](#))

Ithaca, NY

September 2025 – Present

- Engineer remote telemetry channels for an **80+ member team** building hyper-efficient autonomous electric vehicles
- Architected a ROS2 telemetry pipeline over **Starlink** and **Tailscale VPN**, achieving **200 Hz** sensor streaming and **<30 ms** end-to-end latency from shared-memory buffers to a live dashboard, informing decisions for **15+ race engineers**
- Led development of low-power cellular system with AT&T SIM7600 modems, QMI, and Linux kernel networking, cutting vehicle power consumption by **25%** without Starlink while maintaining remote IP connectivity

Bonassar Engineering Lab

Computer Science Research Assistant

Ithaca, NY

September 2025 – May 2026

- Designed a closed-loop PID controller using real time impedance feedback to regulate extrusion for cartilage bioprinting
- Built a visualization GUI to monitor extrusion and motor control in real time, streamlining system testing and calibration
- Implemented and tuned controller in **Python/C++** on a Raspberry Pi Pico, achieving **93%** increased average extruded cell concentration accuracy across **12** experiments, compared to open loop extrusion

ZealGrit

Software Developer (Cornell Hack4Impact)

Ithaca, NY

September 2026 – Present

- Engineering ZealGrit's first digital platform (Express, React Native, Expo, TypeScript) to scale existing **550+ in-person sessions serving 1,850+ women** into a mobile app for feminine health education across rural communities in India
- Designing MongoDB schema for hierarchical content (comics, session modules with quizzes) with Firebase Cloud Storage

PROJECTS

EgoMemory for Meta Ego-Exo4D Recordings | *Python, CLIP, FAISS, PostgreSQL, Tkinter*

[GitHub](#)

- Built a multimodal episodic-memory retrieval system for egocentric video, storing metadata in PostgreSQL and embedding visual frames (CLIP), transcripts, and motion trajectories into a shared vector space for similarity search via FAISS
- Conducted a modality experiment across **50** validated queries to evaluate retrieval performance across different input types
- Multimodal retrieval achieved **84% Recall@5** versus **30%** (vision-only) and **18%** (motion-only), quantifying language and motion's contribution to accurate memory retrieval

CerviCheck Pre-term Birth Diagnosis Device | *Python, SciPy, PySerial, Matplotlib, C++, STM32*

[GitHub](#)

- Extended a Python control GUI to automate pressure sweeps up to **-20 kPa**, integrating impedance-based contact detection to quantify cervical-tissue stress and strain, deformation, and effective modulus for pre-term birth prediction
- Built live visualization and curve fitting, benchmark tests achieved **95% accuracy** in detecting differences in gel thickness
- Ported Arduino C++ pressure-control logic to an STM32 microcontroller; **selected for the e-Lab Startup Accelerator**

TECHNICAL SKILLS

Languages: Java, Python, C++, TypeScript, JavaScript, OCaml, Verilog

Frameworks/Libraries: React, React Native, Expo, Express, FastAPI, NumPy, SciPy, ROS2

Developer Tools: Git, Linux, Docker, PostgreSQL, MongoDB, Firebase, TLA+